CV

Dr. William Ratcliff II

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Education

2003 - Rutgers University, New Brunswick, Ph.D. Physics *Thesis Advisor: Professor Sang W. Cheong*

1997 - University of Michigan, Ann Arbor, BSE in Engineering Physics

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Research Interests

The primary focus of my research has been multiferroic materials. Recently, I have explored geometric ferroelectricity in the LuFeO3 series of compounds in both bulk and thin film samples, culminating in work with collaborators in the development of superlattices with near room temperature magneto-electric effects. Our neutron scattering work was key in understanding the intrinsic magnetic ordering of the LuFeO3 geometric ferroelectric which templates LuFe2O4 in the LuFeO3-LuFe2O4 superlattice to dramatic effect.

Currently, I am working on topological materials. Of particular interest is the interplay of magnetism with topological materials. For example in the MnBi2Te4 system, our measurements of the magnetic structure under applied magnetic fields has direct implications for the band structure. I am also interested in the use of AI to accelerate neutron scattering measurements and analysis.

Career/Experience

2005-Present - Physicist at the NIST Center for Neutron Scattering

2019-Present - Adjunct Associate Professor, Materials Science and Engineering, University of Maryland, College Park

2023-Present - Adjunct Associate Professor, Physics, University of Maryland, College Park

2003-2005 - National Research Council Postdoc at the NIST Center for Neutron Research. Primary work concerned neutron scattering experiments on geometrically frustrated magnets and magnetoelectric systems.

1997-2003 - Worked with Sang W. Cheong in materials physics research including materials synthesis and characterization of novel magnetic systems.

Workshops Organized

- 2026 Short Course: Data Science for Physicists I & II (Chair), APS Global Physics Summit
- 2025 Data Science for Physicists Short Course, APS Global Physics Summit
- 2024 Data Science for Physicists Tutorial, APS Global Physics Summit
- 2023 AI and Scattering Workshop, University of Maryland
- 2023 Data Science Education Community of Practice Summer Workshop, University of Maryland
- 2022 National Society of Black Physicists Conference (Condensed Matter section), University of Virginia
- 2022 PNCMI, Annapolis
- 2022 PNCMI School (organizer/lecturer), NIST, Gaithersburg
- 2021 Data Science Education Community of Practice Workshop
- 2021 School on Representational Analysis and Magnetic Structure (RAMS), University of Maryland
- 2018 ACNS Satellite School: 7th School on Representational Analysis and Magnetic Structure (RAMS)
- 2018 Program Committee International Conference on Magnetism (ICM), San Francisco, CA
- 2017 Co-organizer of Fundamentals of Ferroelectrics Conference (FERRO), Williamsburg, WV
- 2015 5th School on Representational Analysis and Magnetic Structure (RAMS)
- 2012 Workshop on Representational Analysis and Magnetic Structure Determination, BARC, Mumbai, India
- 2011 The NIST/Georgetown 2nd School on Representational Analysis and Magnetic Structures
- 2010 Program Committee MMM Meeting, Atlanta, GA
- 2007 The NIST Workshop on Representational Analysis of Complex Magnetic Structures

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Grants

AIP Venture Grant Data Science Education Community of Practice $100 K for two years (2025) APS Innovation Fund Data Science Education Community of Practice $200,000 for two years (2021)

Selected Honors and Memberships

- **Fellow of American Physical Society**
- Editorial Board, PRX: Intelligence (2026-present)
- APS Committee on Council Committees (January 2026-present)
- Chair APS Topical Group on Magnetism (March 2025 - March 2026)
- Past Chair APS Topical Group on Data Science (March 2024 - March 2025)
- Secretary, Neutron Scattering Society of America (January 2023-present)
- **Associate Editor of Science Advances (2019-present)**
- Neutron Scattering Society of America Service Award (2022)
- NIST Bronze Medal (2021, 2012)
- NIST AI-COI Member at large representing NCNR (2022-present)
- Advisory board CREST Center for Research and Education in Quantum Leap Science and Technology at Norfolk State University (2021)
- Member APS Committee on Honors (January 2023-December 2025)
- APS Mission and Values Working Group (2023-2024)
- Chair NCNR DEIA Committee (2022)
- Member of APS Fellowship Committee (2021-2023)
- Member of APS Committee on Informing the Public (2021-2023)
- Steering Committee APS-IDEA Program (2020-2023)
- Chair SNS-HFIR Beam time review committee single crystal instruments (2020-2023)
- Chapter Leader DataKind DC (2018-2025)
- Elected Chair APS Committee on Minorities in Physics (2017-2018)
- Elected ACA Neutron Scattering Special Interest Group Chair (2015-2016)
- Minority in Research Science Trailblazer award (24th annual BEYA STEM) (2010)
- NRC Postdoctoral Fellowship (2003-2005)
- **Member American Association for the Advancement of Science**
- Member American Physical Society
- Member Sigma Xi
- Member Neutron Scattering Society of America

Publications

Metrics

- **H-index:** 48 (Google Scholar)
- **Citations:** >9000

Book Chapters

1. Multiferroics, William Ratcliff and Jeffrey W. Lynn, Chapter 5 in *Neutron Scattering - Magnetic and Quantum Phenomena*, Felix Fernandez-Alonso and David L. Price, editors (Academic Press, London, 2015).

2. Magnetic Neutron Diffraction: William Ratcliff, Benjamin Frandsen, Juan Rodriguez-Carvajal, Efrain Rodriguez, Clarina Dela Cruz, Chapter 4 in *Powder Diffraction: Theory and Practice 2nd Edition* (Royal Society of Chemistry, 2026)

Publications

104. "Systematics of Neural Networks and Crystal Structure Classification" - Elizabeth Baggett, Edward G. Friedman, Abhishek Shetty, Derrick Chan-Sew, Harshita Dwarcherla, Paul Kienzle, William Ratcliff - *Digital Discovery* (in-progress)

103. "Examining the Spin Structure of Altermagnetic Candidate MnTe Grown with Near Ideal Stoichiometry" - Qihua Zhang, Christopher Jensen, Alexander Grutter, Sandra Santhosh, William Ratcliff, Julie Borchers, Thomas Heitmann, Narendirakumar Narayanan, Timothy Charlton, Mingyu Yu, Ke Wang, Wesley Auker, Nitin Samarth, Stephanie Law - *ACS Nano* (submitted)

102. "Autonomous data collection for the neutron spin echo response function" - Robert Huarcaya, Austin McDannald, William D. Ratcliff, David P. Hoogerheide - *APL Materials* (submitted)

101. "Multiferroicity and phase diagram of ferro-rotational magnet RbFe(SO4)2" - Junjie Yang, Dimuthu Obeysekera, William D. Ratcliff, Lu Li, Sabine N. Neal, Janice L. Musfeldt, Shinichiro Yano - *Journal of Physics: Condensed Matter* **37** (17), 175701 (2025)

100. "Local Inversion Symmetry Breaking and Thermodynamic Evidence for Ferrimagnetism in Fe3GaTe2" - Sang-Eon Lee, Yue Li, Yeonkyu Lee, W. Kice Brown, PeiYu Cai, Jinyoung Yun, Chanyoung Lee, Alex Moon, Lingrui Mei, Jaeyong Kim, Yan Xin, Julie A. Borchers, Thomas W. Heitmann, Matthias Frontzek, William D. Ratcliff, Gregory T. McCandless, Julia Y. Chan, Elton J. G. Santos, Jeehoon Kim, Charudatta M. Phatak, Vadym Kulichenko, Luis Balicas - *ACS Nano* **19** (31), 28702–28718 (2025)

99. "Topological Hall effect induced by chiral fluctuations in ErMn6Sn6" - Kyle Fruhling, Alenna Streeter, Sougata Mardanya, Xiaoping Wang, Priya Baral,

Oksana Zaharko, Igor I. Mazin, Sugata Chowdhury, William D. Ratcliff, Fazel Tafti - *Physical Review Materials* **8** (9), 094411 (2024)

98. “Data science education in undergraduate physics: Lessons learned from a community of practice” - K. Shah, J. Butler, A.V. Knaub, A. Zenginoğlu, W. Ratcliff, M. Soltanieh-Ha - *American Journal of Physics* **92** (9), 655-662 (2024)

97. “Electric-Field Manipulation of Magnetic Chirality in a Homo-Ferro-Rotational Helimagnet” - Junjie Yang, Masaaki Matsuda, Trevor Tyson, Joshua Young, William Ratcliff, Yunpeng Gao, Dimuthu Obeysekera, Xiaoyu Guo, Rachel Owen, Liuyan Zhao, Sang-wook Cheong - *Advanced Science* **11** (33), 2402048 (2024)

96. “Effects of dimensionality on the electronic structure of Ruddlesden-Popper chromates Srn+1CrnO3n+1” - Spencer Doyle, Lerato Takana, Margaret A. Anderson, Dan Ferenc Segedin, Hesham El-Sherif, Charles M. Brooks, Xiaoping Wang, Padraic Shafer, Alpha T. N’Diaye, Ismail El Baggari, William D. Ratcliff, Andrés Cano, Quintin N. Meier, Julia A. Mundy - *Physical Review Materials* **8** (7), L071602 (2024)

95. “Observation of Deuterated Double-Perovskite Hydroxide CoSn(OH)6 Nanocubes” - Menuka Adhikari, Starfari T. McClain, Rekha George, Sivasankara Rao Ede, Hui Wu, William D. Ratcliff, Liurukara Sanjeewa, Cheng Li, Zhiping Luo - *Microscopy and Microanalysis* **29** (Supplement_1), 1354-1355 (2023)

94. “Antiferromagnetic metal phase in an electron-doped rare-earth nickelate” - Qi Song, Spencer Doyle, Grace A. Pan, Ismail El Baggari, Dan Ferenc Segedin, Denisse Córdova Carrizales, Johanna Nordlander, Christian Tzschaschel, James R. Ehrets, Zubia Hasan, Hesham El-Sherif, Jyoti Krishna, Chase Hanson, Harrison LaBollita, Aaron Bostwick, Chris Jozwiak, Eli Rotenberg, Su-Yang Xu, Alessandra Lanzara, Alpha T. N’Diaye, Colin A. Heikes, Yaohua Liu, Hanjong Paik, Charles M. Brooks, Betül Pamuk, John T. Heron, Padraic Shafer, William D. Ratcliff, Antia S. Botana, Luca Moreschini, Julia A. Mundy - *Nature Physics* **19**, 522 (2023)

93. “ANDiE the Autonomous Neutron Diffraction Explorer” - Austin McDannald, Matthias Frontzek, Andrei T. Savici, Mathieu Doucet, Efrain E. Rodriguez, Kate Meuse, Jessica Opsahl-Ong, Daniel Samarov, Ichiro Takeuchi, William Ratcliff, A. Gilad Kusne - *Neutron News* **34** (2), 6-7 (2023)

92. “Fast broadband cluster spin-glass dynamics in PbFe1/2Nb1/2O3” - C. Stock, B. Roessli, P. M. Gehring, J. A. Rodriguez-Rivera, N. Giles-Donovan, S. Cochran, G. Xu, P. Manuel, M. J. Gutmann, W. D. Ratcliff, T. Fennell, Y. Su, X. Li, and H. Luo - *Phys. Rev. B* **106**, 144207 (2022)

91. “On-the-fly Autonomous Control of Neutron Diffraction via Physics-Informed Bayesian Active Learning” - Austin McDannald, Matthias Frontzek, Andrei Savici, Mathieu Doucet, Efrain Rodriguez, Kate Meuse, Jessica Opsahl-Ong, Daniel Samarov, Ichiro Takeuchi, William Ratcliff, Aaron G. Kusne - *Applied Physics Reviews* **9**, 021408 (2022)

90. “A Semi-supervised deep-learning based classification of neutron diffraction data into space groups” - Satvik Lolla, Haotong Ling, Ichiro Takeuchi, Aaron G. Kusne, William Ratcliff - *Journal of Applied Crystallography* **55** (2022)

89. “Revealing the Symmetry of Materials through Neutron Diffraction” - W. Ratcliff - *Symmetry* **14**, 1215 (2022)

88. “Liberating a hidden antiferroelectric phase with interfacial electrostatic engineering” - Julia A. Mundy, Colin A. Heikes, Bastien F. Grosso, Dan Ferenc Segedin, Zhe Wang, Yu-Tsen Shao, Cheng Dai, Berit H. Goodge, Quintin N. Meier, Christopher T. Nelson, Bhagwati Prasad, Fei Xue, Lena F. Kourkoutis, Long-Qing Chen, William D. Ratcliff, Nicola A. Spaldin, Ramamoorthy Ramesh, Darrell G. Schlom - *Science Advances* **8**, DOI: 10.1126/sciadv.abg5860 (2022)

87. “Informal Science Education and Career Advancement” - Michael Smith, Claudia Fracchiolla, Sean Fleming, Arturo Dominguez, Alexandra Lau, Shannon Greco, Don Lincoln, Eleni Katifori, William Ratcliff, Maria Longobardi, Maajida Murdock, Mustapha Ishak - arXiv:2112.10623 (2021)

86. “Magnetic field-induced non-trivial electronic topology in FeGeTe2” - Juan Macy, Danilo Ratkovski, Purnima P. Balakrishnan, Mara Strungaru, Yu-Che Chiu, Aikaterini Flessa, Alex Moon, Wenkai Zheng, Ashley Weiland, Gregory T. McCandless, Julia Y. Chan, Govind S. Kumar, Michael Shatruk, Alexander J. Grutter, Julie A. Borchers, William D. Ratcliff, Eun Sang Choi, Elton JG Santos, Luis Balicas - *Applied Physics Reviews* **8**, 041401 (2021)

85. “Synthesis and Characterization of Sr2Co2-xFexO5+d Perovskite Oxides” - Sivasankara Rao Ede, Carlos Poasada, Jessa Guffie, William Ratcliff, Hui Wu, Shubo Han, and Zhiping Luo - *Microsc. Microanal.* **27** (Suppl 1) (2021)

84. “Magnetic ordering and structural distortion in a PrFeAsO single crystal studied by neutron and x-ray scattering” - M.G. Kim, W. Ratcliff, D.M. Pajerowski, J-W. Kim, J-Q. Yan, J.W. Lynn, A.I. Goldman, A. Kreyssig - *Phys. Rev. B* **103**, 174405 (2021)

83. “Intermediate Sr2Co1.5Fe0.5O6-delta Tetragonal Structure between Perovskite and Brownmillerite as a Model Catalyst with Layered Oxygen Deficiency for Enhanced Electrochemical Water Oxidation” - Sivasankara Rao Ede, Candyce Collins, Carlos Posada, Gibin George, Hui Wu, William D. Ratcliff, Yulin Lin, Jianguo Wen, Shubo Han, and Zhiping Luo - *ACS Catal.* **11**, 4327 (2021)

82. “Control of magnetoelectric coupling in the Co2Y-type hexaferrites” - Chang Bae Park, Kwang Woo Shin, Sae Hwan Chun, Jun Han Lee, Yoon Seok Oh, Steven M. Disseler, Colin A. Heikes, William D. Ratcliff, Woo-Suk Noh, Jae-Hoon Park, Kee Hoon Kim - *Phys. Rev. Materials* **5**, 034412 (2021)

81. “Kitaev interactions in Co honeycomb antiferromagnets Na3Co2SbO6 and Na2Co2TeO6” - M. Songvilay, J. Robert, J. A. Rodriguez-Rivera, W. D. Ratcliff, F. Damay, V. Baledent, M. Jimenez-Ruiz, P. Lejay, E. Pachoud, A. Hadj-Azzem, V. Simonet, and C. Stock - *Phys. Rev. B* **102**, 224429 (2020)

80. "Evolution of the magnetic properties in simultaneously Cd and Ir-doped Ce2RhIn8 antiferromagnet" - D. S. Christovam, G. S. Freitas, M. M. Piva, J. C. Souza, M. O. Malcolms, J. Leao, W. Ratcliff, J. W. Lynn, C. Adriano, P. G. Pagliuso - *Phys. Rev. B* **102**, 195137 (2020)

79. "Large exchange splitting in monolayer graphene magnetized by an antiferromagnet" - Yingying Wu, Gen Yin, Lei Pan, Alexander J. Grutter, Quanjun Pan, Albert Lee, Eun Sang Choi, Mingliang Tian, Peng Deng, Qiming Shao, Shin-Hung Tsai, Qinglin He, Dustin A. Gilbert, Julie A. Borchers, William Ratcliff II, Ang Li, Xiao-dong Han, and Kang L. Wang - *Nature Electronics* (2020)

78. "Magnetic phase transitions and spin density distribution in the molecular multiferroic GaV4S8 system" - Rebecca L. Dally, William D. Ratcliff II, Lunyong Zhang, Heung-Sik Kim, Markus Bleul, J.W. Kim, Kristjan Haule, David Vanderbilt, Sang-Wook Cheong, and Jeffrey W. Lynn - *Phys. Rev. B* **102**, 014410 (2020)

77. "Termination switching of antiferromagnetic proximity effect in topological insulator" - Chao-Yao Yang, Lei Pan, Alexander J. Grutter, Haiying Wang, Xiaoyu Che, Qing Lin, Yingying Wu, D. A. Gilbert, Padraic Shafer, Elke Arenholz, Hao Wu, Gen Yin, Peng Deng, J. A. Borchers, W. Ratcliff II, and Kang L. Wang - *Science Advances* **6**, eaaz8463 (2020)

76. "Quantum oscillations from networked topological interfaces in a Weyl semimetal" - I-Lin Liu, Colin Heikes, Taner Yildirim, Chris Eckberg, Tristin Metz, ShengRan, William Ratcliff II, Johnpierre Paglione, and Nicholas P. Butch - *Nature Quantum Materials* **5**, Article 62 (2020)

75. "Ferromagnetic van der Waals compound MnSb1.8Bi0.2Te4" - Yangyang Chen, Ya-Wen Chuang, Seng Huat Lee, Yanglin Zhu, Kevin Honz, Yingdong Guan, Yu Wang, Ke Wang, Zhiqiang Mao, Colin Heikes, P. Quarterman, Pawel Zajdel, Julie A. Borchers, William Ratcliff II, Jun Zhu - *Phys. Rev. Matt.* **4**, 064411 (2020)

74. "Electronic and magnetic properties of stoichiometric CeAuBi2" - M. M. Piva, R. Tartaglia, G. S. Freitas, J. C. Souza, D. S. Christovam, S. M. Thomas, J. Leao, W. Ratcliff, J. W. Lynn, C. Lane, J.-X. Zhu, J. D. Thompson, P. F. S. Rosa, C. Adriano, E. Granado, and P. G. Pagliuso - *Phys. Rev. B* **101**, 214431 (2020)

73. "Common acoustic phonon lifetimes in inorganic and hybrid lead halide perovskites" - M. Songvilay, N. Giles-Donovan, M. Bari, Z-G. Ye, J.L. Minns, M.A. Green, Guangyong Xu, P.M. Gehring, K. Schmalzl, W.D. Ratcliff, C.M. Brown, D. Chernyshov, W. van Beek, S. Cochran, C. Stock - *Physical Review Materials* **3**, 093602 (2019)

72. "Exchange Bias in Bulk alpha-Fe / gamma-Fe70Mn30 Nanocomposites for Permanent Magnet Applications" - Ian J. McDonald, Michelle Jamer, Kathryn L. Krycka, Elaf Anber, Daniel Foley, Andrew Charles Lang, William Ratcliff,

Don Heiman, Mitra L. Taheri, Julie A. Borchers, Laura H. Lewis - *ACS Appl. Nano Mater.* **2**, 1940 (2019)

71. “Spin Scattering and Noncollinear Spin Structure-Induced Intrinsic Anomalous Hall effect in Antiferromagnetic Topological Insulator MnBi2Te4” - Seng Huat Lee, Yanglin Zhu, Yu Wang, Leixin Miao, Hemian Yi, Timothy Pillsbury, Susan Kempinger, Jin Hu, Colin A. Heikes, Patrick A. Quarterman, William D. Ratcliff, Julie A. Borchers, H. Zhang, Xianglin Ke, David Graf, Nasim Alem, Cui-Zu Chang, Nitin Samarth, and Zhiqiang Mao - *Phys. Rev. Research* **1**, 012011(R) (2019)

70. “Probing Superexchange Interactions in Spin Ice Ho2Ti2O7 Thin Films” - Kevin Barry, Naween Anand, Biwen Zhang, Yan Xin, Arturas Vailionis, Colin Heikes, Haidong Zhou, Y. Qiu, William Ratcliff, Christianne Beekman - *Phys. Rev. Materials* **3**, 084412 (2019)

69. “Fe2MnGe: A Hexagonal Heusler Analogue” - S. Keshavarz, N. Naghibolashra, M.E. Jamer, K. Vinson, D. Mazumdar, C.L. Dennis, W. Ratcliff II, J.A. Borchers, A. Gupta, P. LeClair - *Journal of Alloys and Compounds* **771**, 793 (2019)

68. “Spin Rotation induced by applied pressure in Cd-doped Ce2RhIn8 intermetallic compound” - D. S. Christovam, C. Giles, L. Mendonca-Ferreira, J. Leao, W. Ratcliff, J. W. Lynn, S. Ramos, E. N. Hering, H. Hidaka, E. Baggio-Saitovich, Z. Fisk, P. G. Pagliuso, C. Adriano - *Physical Review B* **100**, 165133 (2019)

67. “Mechanical control of crystal symmetry and superconductivity in Weyl semimetal MoTe2” - Colin Heikes, I-Lin Liu, Tristin Metz, Chris Eckberg, Paul Neves, Yan Wu, Linda Hung, Phil Piccoli, Huibo Cao, Juscelino Leao, Johnpierre Paglione, Taner Yildirim, Nicholas P. Butch, William Ratcliff II - *Phys. Rev. Materials* **2**, 074202 (2018)

66. “reductus: a stateless Python data-reduction service with a browser frontend” - Brian Maranville, William Ratcliff II, Paul Kienzle - *Journal of Applied Crystallography* **51**, 1500 (2018)

65. “Ordered Magnetism in the Decorated 3D Ising jeff = 1/2 alpha-CoV3O8” - P. M. Sarte, A. M. Arevalo-Lopez, M. Songvilay, D. Le, T. Guidi, V. Garcia-Sakai, S. Mukhopadhyay, S. C. Capelli, W. D. Ratcliff, K. H. Hong, G. M. McNally, E. Pachoud, J. P. Attfield, and C. Stock - *Phys. Rev. B* **98**, 224410 (2018)

64. “Lifetime-shortened acoustic phonons and static order at the Brillouin zone boundary in the organic-inorganic perovskite CH3NH3PbCl3” - M. Songvilay, C. Stock, Z.-G. Ye, Guangyong Xu, P. M. Gehring, W. D. Ratcliff, K. Schmalzl, F. Bourdarot, and B. Roessli - *Phys. Rev. Materials* **2**, 123601 (2018)

62. “Electric-field Induced Reversible Switching of the Magnetic Easy-axis in Co/BiFeO3 on SrTiO3” - Tieren Gao, Xiaohang Zhang, William Ratcliff, Shingo Maruyama, Makoto Murakami, Anbusathaiah Varatharajan, Zahra Yamani, Peijie Chen, Ke Wang, Huairuo Zhang, Robert D. Shull, Leonid A. Bendersky,

John Unguris, Ramamoorthy Ramesh, Ichiro Takeuchi - *Nano Letters* **17**, 2825 (2017)

61. “Successive field-induced transitions in BiFeO3 around room temperature” - Shiro Kawachi, Atsushi Miyake, Toshimitsu Ito, Sachith E. Dissanayake, Masaaki Matsuda, W. Ratcliff, Zhijun Xu, Yang Zhao, Shin Miyahara, Nobuo Furukawa, Masashi Tokunaga - *Phys. Rev. Materials* **1**, 024408 (2017)

60. “Structural and Magnetic Phase Transitions in Chromium Nitride Thin Films Grown by RF Nitrogen Plasma Molecular Beam Epitaxy” - Khan Alam, Steven M. Disseler, William D. Ratcliff, Julie A. Borchers, Rodrigo Ponce-Perez, Gregorio H. Cocoletzi, Noboru Takeuchi, Andrew Foley, Andrea Richard, David C. Ingram, Arthur R. Smith - *Phys. Rev. B* **96**, 104433 (2017)

59. “Pressure tuning of collapse of helimagnetic structure in Au2Mn” - I-Lin Liu, Maria J. Pascale, Juscelino B. Leao, Craig M. Brown, William D. Ratcliff, Qingzhen Huang, and Nicholas P. Butch - *Phys. Rev. B* **96**, 184429 (2017)

58. “Tailoring Exchange Couplings in Magnetic Topological Insulator/Antiferromagnet Heterostructures” - Qing Lin He, Xufeng Kou, Alexander J. Grutter, Gen Yin, Lei Pan, Xiaoyu Che, Yuxiang Liu, Tianxiao Nie, Bin Zhang, Steven M. Disseler, Brian J. Kirby, William Ratcliff II, Qiming Shao, Koichi Murata, Xiaodan Zhu, Guoqiang Yu, Yabin Fan, Mohammad Montazeri, Xiaodong Han, Julie A. Borchers, Kang L. Wang - *Nature Materials* **16**, 94 (2017)

57. “Magnetic Structures and Dynamics of Multiferroic Systems Obtained with Neutron Scattering” - W. D. Ratcliff, II, Jeffrey W. Lynn, Valery Kiryukhin, Prashant Jain, and Michael R. Fitzsimmons - *Nature Partner Journals: Quantum Materials* **1**, 16003 (2016)

56. “Atomically engineered ferroic layers yield a room temperature magnetoelectric multiferroic” - Julia A. Mundy, Charles M. Brooks, Megan E. Holtz, Jarrett A. Moyer, Hena Das, Alejandro F. Rbola, John T. Heron, James D. Clarkson, Steven M. Disseler, Zhiqi Liu, Alan Farhan, Rainer Held, Robert Hovden, Elliot Padgett, Qingyun Mao, Hanjong Paik, Rajiv Misra, Lena F. Kourkoutis, Elke Arenholz, Andreas Scholl, Julie A. Borchers, William D. Ratcliff, Ramamoorthy Ramesh, Craig J. Fennie, Peter Schiffer, David A. Muller, Darrell G. Schlom - *Nature* **537**, 523 (2016)

55. “Bayesian method for the analysis of diffraction patterns using BLAND” - J.E. Lesniewski, S.M. Disseler, D.J. Quintana, P.A. Kienzle, W.D. Ratcliff - *Journal of Applied Crystallography* **49**, 2201-2209 (2016)

54. “Magnetic Structure and Ordering of Multiferroic Hexagonal LuFeO3” - Steven M. Disseler, Julie A. Borchers, Charles M. Brooks, Julia A. Mundy, Jarrett A. Moyer, Daniel A. Hillsberry, Eric L. Thies, Dmitri A. Tenne, John Heron, Megan E. Holtz, James D. Clarkson, Gregory M. Stiehl, Peter Schiffer, David A. Muller, Darrell G. Schlom, William D. Ratcliff - *Physical Review Letters* **114**, 217602 (2015)

53. "Multiferroicity in doped hexagonal LuFeO3" - Steven M. Disseler, Xuan Luo, Bin Gao, Yoon Seok Oh, Rongwei Hu, Yazhong Wang, Dylan Quintana, Alexander Zhang, Qingzhen Huang, June Lau, Rick Paul, Jeffrey W. Lynn, Sang-Wook Cheong, and William Ratcliff, II - *Phys. Rev. B* **92**, 054435 (2015)

52. "One Dimensional(1D)-to-2D Crossover of Spin Correlations in the 3D Magnet ZnMn2O4" - S. M. Disseler, Y. Chen, S. Yeo, G. Gasparovic, P. M. B. Piccoli, A. J. Schultz, Y. Qiu, Q. Huang, S-W. Cheong, and W. Ratcliff II - *Scientific Reports* **5**, 17771 (2015)

51. "Complex structures of different CaFe2As2 samples" - B. Saparov, C. Cantoni, M.H. Pan, T.C. Hogan, W. Ratcliff, S.D. Wilson, K. Fritsch, B.D. Gaulin, A.S. Sefat - *Scientific Reports*, 4120 (2014)

50. "Reflections on the magnetic pair distribution function" - W. Ratcliff - *Acta Crystallographica A* **70**, 1 (2014)

49. "Change in the magnetic structure of (Bi,Sm)FeO3 thin films at the morphotropic phase boundary probed by neutron diffraction" - Shingo Maruyama, Varatharajan Anbusathaiah, Amy Fennell, Mechthild Enderle, Ichiro Takeuchi, William D. Ratcliff - *APL Materials* **2**, doi:10.1063/1.4901294 (2014)

48. "Electric-field-controlled antiferromagnetic domains in epitaxial BiFeO3 thin films probed by neutron diffraction" - W. Ratcliff II, Zahra Yamani, V. Anbusathaiah, T.R. Gao, P.A. Kienzle, H. Cao, I. Takeuchi - *Phys. Rev. B* **87**, 140405 (2013)

47. "Double Focusing Thermal Triple-Axis Spectrometer at the NCNR" - Lynn, J.W., Chen, Y., Chang, S., Zhao, Y., Chi, S., W. Ratcliff, II, Ueland, B.G., Erwin, R.W. - *Journal of Research of the National Institute of Standards and Technology* **117**, 61 (2012)

46. "Local Weak ferromagnetism in single-crystalline ferroelectric BiFeO3" - M. Ramazanoglu, M. Laver, W. Ratcliff II, S.M. Watson, W.C. Chen, A. Jackson, K. Kothapalli, Seongsu Lee, S.-W. Cheong, V. Kiryukhin - *Phys. Rev. Lett.* **107**, 207206 (2011)

45. "Antiferromagnetic order and superlattice structure in nonsuperconducting and superconducting Rb(y)Fe(1.6+x)Se2" - *Phys. Rev. B* **84**, 094504 (2011)

44. "Giant Effect of Uniaxial Pressure on Magnetic Domain Populations in Multiferroic Bismuth Ferrite" - M. Ramazanoglu, W. Ratcliff, H.T. Yi, A.A. Sirenko, S.W. Cheong, V. Kiryukhin - *Phys. Rev. Lett.* **107**, 067203 (2011)

43. "Temperature-dependent properties of the magnetic order in single-crystal BiFeO3" - M. Ramazanoglu, W. Ratcliff II, Y.J. Choi, Seongsu Lee, S-W. Cheong, V. Kiryukhin - *Phys. Rev. B* **83**, 174434 (2011)

41. "Mechanism of exchange-striction of ferroelectricity in multiferroic orthorhombic HoMnO3 single crystals" - N. Lee, Y.J. Choi, M. Ramazanoglu, W. Ratcliff II, V. Kiryukhin, and S-W. Cheong - *Phys. Rev. B* **84**, 020101 (2011)

40. "Neutron Diffraction Investigations of Magnetism in BiFeO3 Epitaxial Films" - William Ratcliff II, Daisuke Kan, Wangchun Chen, Shannon Watson, Songxue Chi, Ross Erwin, Garry J. McIntyre, Sylvia C. Capelli, Ichiro Takeuchi - *Advanced Functional Materials* **21**, 1567 (2011)

39. "Antiferromagnetic Order and Superlattice Structure in Nonsuperconducting and Superconducting RbyFe1.6+xSe2" - Wang, M. et al. - *Physical Review B* **84**, 094504-1 (2011)

38. "Nitrogen contamination in elastic neutron scattering" - Songxue Chi, Jeffrey W. Lynn, Ying Chen, William Ratcliff II, Benjamin G. Ueland, Nicholas P. Butch, Shanta R. Saha, Kevin Kirshenbaum, Johnpierre Paglione - *Measurement Science and Technology* **22**, 047001 (2011)

37. "Incommensurate Magnetism in FeAs Strips: Neutron Scattering from CaFe4As3" - Yusuke Nambu, Liang L. Zhao, Emilia Morosan, Kyoo Kim, Gabriel Kotliar, Pawel Zajdel, Mark A. Green, William Ratcliff, Jose A. Rodriguez-Rivera, Collin Broholm - *Phys. Rev. Lett.* **106**, 037201 (2011)

36. "Interplay of Fe and Nd magnetism in NdFeAsO single crystals" - W. Tian, W. Ratcliff II, M. G. Kim, J.-Q. Yan, P.A. Kienzle, Q. Huang, B. Jensen, K.W. Dennis, R.W. McCallum, T.A. Lograsso, R.J. McQueeney, A.I. Goldman, J.W. Lynn, A. Kreyssig - *Phys. Rev. B* **82**, 060514 (2010)

35. "Cd Doping Effects in the heavy-fermion compounds Ce2Min8 (M=Rh and Ir)" - C. Adriano, C. Giles, E.M. Bittar, L.N. Coelho, F. De Bergevin, C. Mazzoli, L. Paolasini, W. Ratcliff, R. Bindel, J.W. Lynn, Z. Fisk, P.G. Pagliuso - *Phys. Rev. B* **81**, 245115 (2010)

34. "Magnetic form factor of SrFe2As2: Neutron diffraction measurements" - W. Ratcliff II, P.A. Kienzle, Jeffrey W. Lynn, Shiliang Li, Pengcheng Dai, G.F. Chen, N.L. Wang - *Phys. Rev. B* **81**, 140502 (2010)

33. "Evolution of the bulk properties, structure, magnetic order, and superconductivity with Ni doping in CaFe(2-x)NixAs2" - N. Kumar, S.X. Chi, Y. Chen, K.G. Rana, A.K. Nigam, A. Thamizhavel, W. Ratcliff, S.K. Dar, J.W. Lynn - *Phys. Rev. B* **80**, 144524 (2009)

32. "Crossover from incommensurate to commensurate magnetic orderings in CoCr2O4" - L.J. Chang, D.J. Huang, W.H. Li, S.W. Cheong, W. Ratcliff, J.W. Lynn - *J. Phys. Cond. Matt.* **21**, 456008 (2009)

31. "Short Range Incommensurate magnetic order near the superconducting phase boundary in Fe(1+delta)Te(1-x)Sex" - J.S. Wen, G.Y. Xu, Z.J. Xu, Z.W. Lin, Q. Li, W. Ratcliff, G. Gu, J.M. Tranquada - *Phys. Rev. B* **80**, 104506 (2009)

30. "Spin-Lattice Order in Frustrated ZnCr2O4" - S. Ji, S.H. Lee, C. Broholm, T.Y. Koo, W. Ratcliff, S.W. Cheong, P. Zschack - *Phys. Rev. Lett.* **103**, 037201 (2009)

29. “The Magnetic ground state of CaMn2Sb2” - W. Ratcliff II, A.L.L. Sharma, A.M. Gomes, J.L. Gonzalez, Q. Huang, J. Singleton - *J. Mag. Mag. Materials* **321**, 2612 (2009)

28. “Order by Static Disorder in the Ising Chain Magnet Ca3Co2-xMnxO6” - V. Kiryukhin, S. Lee, W. Ratcliff II, Q. Huang, H.T. Yi, Y.J. Choi, S.W. Cheong - *Phys. Rev. Lett.* **102**, 187202 (2009)

27. “3:1 magnetization plateau and suppression of ferroelectric polarization in an Ising chain multiferroic” - Y.J. Jo, S. Lee, E.S. Choi, H.T. Yi, W. Ratcliff II, Y.J. Choi, V. Kiryukhin, S.W. Cheong, L. Balicas - *Phys. Rev. B* **79**, 012407 (2009)

26. “Low energy spin waves and magnetic interactions in SrFe2As2” - Jun Zhao, Dao Xin Yao, Shiliang Li, Tao Hong, Y. Chen, S. Chang, W. Ratcliff II, J. W. Lynn, H. A. Mook, G. F. Chen, J. L. Luo, N. L. Wang, E. W. Carlson, Jiangping Hu, and Pengcheng Dai - *Phys. Rev. Lett.* **101**, 167203 (2008)

25. “Spin and Lattice Structure of Single Crystal SrFe2As2” - Jun Zhao, W. Ratcliff II, J-W. Lynn, G.F. Chen, J.L. Luo, N.L. Wang, Jiangping Hu, Pengcheng Dai - *Phys. Rev. B* **78**, 140504 (2008)

24. “Magnetic order close to superconductivity in the iron-based layered LaO1-xFxFeAs systems” - C. dela Cruz, Q. Huang, J.W. Lynn, J.Y. Li, W. Ratcliff, J.L. Zaretsky, H.A. Mook, G.F. Chen, J.L. Luo, N.L. Wang, P.C. Dai - *Nature* **453**, 899 (2008)

23. “Neel to Spin-Glass-Like Transition Versus Dilution in Geometrically Frustrated ZnCr2-2xGa2xO4” - S-H. Lee, W. Ratcliff, Q. Huang, T.H. Kim, S-W. Cheong - *PRB* **77**, 014405 (2008)

22. “Formation of pancakelike ising domains and giant magnetic coercivity in ferrimagnetic LuFe2O4” - W. Wu, V. Kiryukhin, H.-J. Noh, K.-T. Ko, J.-H. Park, W. Ratcliff II, P. A. Sharma, N. Harrison, Y.J. Choi, Y. Horime, S. Lee, S. Park, H.T. Yi, C.L. Zhang, S-W. Cheong - *PRL* **101**, 137203 (2008)

21. “Electric field control of the magnetic state in BiFeO3 single crystals” - Seongsu Lee, W. Ratcliff, S.W. Cheong, V. Kiryukhin - *APL* **92**, 192906 (2008)

20. “Single Ferroelectric and chiral magnetic domain of single-crystalline BiFeO3 in an electric field” - Seongsu Lee, Taekjib Choi, W. Ratcliff II, R. Erwin, S-W. Cheong, and V. Kiryukhin - *PRB RC* **78**, 100101 (2008)

19. “The pressure effect on the magnetic commensurability and ferroelectricity in multiferroic HoMn2O5” - C.R. dela Cruz, B. Lorenz, W. Ratcliff, J. Lynn, M.M. Gospodinov, C.W. Chu - *Physica B* **403**, 1359 (2008)

18. “Observation of a continuous phase transition in a shape-memory alloy” - J.C. Lashley, S.M. Shapiro, B.L. Winn, C.P. Opeil, M.E. Manley, A. Alatas, W. Ratcliff, T. Park, R.A. Fisher, B. Mihaila, P. Riseborough, E.K.H. Salje, J.L. Smith - *Phys. Rev. Lett.* **101**, 135703 (2008)

17. “Crystal distortions in geometrically frustrated ACr2O4 (A=Zn,Cd)” - S.-H. Lee, G. Gasparovic, C. Broholm, M. Matsuda, J-H. Chung, Y.J. Kim, H. Ueda, G. Xu, P. Zschack, K. Kakurai, H. Takagi, W. Ratcliff, T-H. Kim, S-W. Cheong - *Journal of Physics Condensed Matter* **19**, 145259 (2007)

16. “Evidence for strong spin-lattice coupling in multiferroic RMn2O5 (R=Tb, Dy, Ho) via thermal expansion anomalies” - C.R. dela Cruz, F. Yen, B. Lorenz, S. Park, S.W. Cheong, M.M. Gospodinov, W. Ratcliff, J.W. Lynn, C.W. Chu - *Journal of Applied Physics* **99**, 08R2103 (2006)

15. “Structural Anomalies at the magnetic and ferroelectric transitions in RMn2O5 (R=Tb, Dy, Ho)” - C.R. dela Cruz, F. Yen, B. Lorenz, M.M. Gospodinov, C.W. Chu, W. Ratcliff, J.W. Lynn, S. Park, S-W. Cheong - *Phys. Rev. B* **73**, 100406 (2006)

14. “Conformation of the HIV-1 Gag protein in solution” - S.A.K. Datta, J.E. Curtis, W. Ratcliff, P.K. Clark, R.M. Crist, J. Lebowitz, S. Krueger, A. Rein - *Journal of Molecular Biology* **365**, 812 (2007)

13. “Structural anomalies at the magnetic and ferroelastic transitions in RMn2O5 (R=Tb, Dy, Ho)” - C.R. dela Cruz, F. Yen, B. Lorenz, M.M. Gospodinov, C.W. Chu, W. Ratcliff, J.W. Lynn, S. Park, and S-W. Cheong - *Phys. Rev. B* **73**, 100406 (2006)

12. “Pressure-dependent magnetic properties of geometrically frustrated ZnCr2O4” - Y. Jo, J.G. Park, H.C. Kim, W. Ratcliff, S-W. Cheong - *Phys. Rev. B* **72**, 184421 (2005)

11. “Magnetic Phase Diagram of the Colossal Magnetoelectric, DyMn2O5” - W. Ratcliff II, V. Kiryukhin, M.A. Kenzelmann, S-H. Lee, J. Schefer, R. Erwin, N. Hur, S. Park, and S-W. Cheong - *Phys. Rev. B (R)* **72**, 060407 (2005)

10. “Effects of spin-phonon coupling in the magnetic transition in strongly frustrated ZnCr2O4” - A. B. Sushkov, H. D. Drew, O. Tchernyshyov, W. Ratcliff, S.W. Cheong - *Phys. Rev. Lett.* **94**, 137202 (2005)

9. “Spin Singlet formation in MgTi2O4: evidence of a chiral dimerization pattern” - M. Schmidt, W. Ratcliff II, P.G. Radaelli, K. Refson, N.M. Harrison, S.W. Cheong - *Phys. Rev. Lett.* **92**, 056402 (2004)

8. “First-Order nature of the ferromagnetic phase transition in (La-Ca)MnO3 near optimal doping” - C.P. Adams, J.W. Lynn, V.N. Smolyaninova, A. Biswas, R.L. Green, W. Ratcliff II, S.-W. Cheong, Y.M. Mukovskii, D.A. Shulyatev - *Phys. Rev. B* **70**, 134414 (2004)

7. “Emergent excitations in a geometrically frustrated magnet” - S.-H. Lee, C. Broholm, W. Ratcliff, G. Gasparovic, Q. Huang, T.H. Kim and S-W. Cheong - *Nature* **418**, 856-858 (2002)

6. “Fluctuations and Freezing of Spin-Correlated Nanoclusters in a Geometrically Frustrated Magnet” - W. Ratcliff II, S.-H. Lee, C. Broholm, S-W. Cheong, Q.

Huang - *Physical Review B: Rapid Communications* **65**, 220406/1-220406/4 (2002)

5. “Magnetic properties of the frustrated antiferromagnetic spinel ZnCr2O4 and the spin glass Zn1-xCdxCr2O4 (x=0.05,0.10)” - H. Martinho, N.O. Moreno, J.A. Sanjurjo, C. Rettori, A.J. Garcia-Adeva, D.L. Huber, S.B. Oseroff, W. Ratcliff, S.W. Cheong, P.G. Pagliuso, J.L. Sarrao, G.B. Martins - *Phys. Rev. B* **64**, 024408/1-024408/6 (2001)

4. “Studies of the three-dimensional frustrated antiferromagnetic ZnCr2O4” - H. Martinho, N.O. Moreno, J.A. Sanjurjo, C. Rettori, A.J. Garcia-Adeva, D.L. Huber, S.B. Oseroff, W. Ratcliff, S.W. Cheong, P.G. Pagliuso, J.L. Sarrao, G.B. Martins - *Journal of Applied Physics* **89**, 7050-7052 (2001)

3. “Muon spin relaxation study of La1-xCaxMnO3” - R.H. Heffner, J.E. Sonier, D.E. MacLaughlin, G.J. Nieuwenhuys, G.M. Luke, Y.J. Uemura, W. Ratcliff, S.W. Cheong, G. Balakrishnan - *Phys. Rev. B* **63**, 094408/1-094408/14 (2001)

2. “Local spin resonance and spin-Peierls-like phase transition in a geometrically frustrated antiferromagnet” - S.H. Lee, C. Broholm, T.H. Kim, W. Ratcliff, S.W. Cheong - *Phys. Rev. Lett.* **84**, 3718-3721 (2000)

1. “Intergrain magnetoresistance via second-order tunneling in perovskite manganites” - S. Lee, H.Y. Hwang, B.I. Shraiman, W.D. Ratcliff, S.W. Cheong - *Phys. Rev. Lett.* **82**, 4508-4511 (1999)

Selected Conference Talks/Posters

2026

- “AI and Neutron Scattering”, Gordon Research Conference on Multifunctional Materials and Structures: Enabling Multifunctionality through AI Informed Material Discovery and System Design, Ventura, CA (invited)
- “Introduction to Neural Networks”, Short Course: Data Science for Physicists I, APS Global Physics Summit, Denver, CO (invited)
- “AI and Neutron Scattering”, Department of Physics, George Mason University (April 2026) (invited)
- “Attention is not all you need: Comparing the performance of transformers and CNNs in classifying space groups from powder diffraction data”, APS Global Physics Summit, Denver, CO

2025

- “Single Crystal Diffraction”, 1st Neutron Scattering School (invited)
- “Accelerating Materials Discovery: The Role of AI in Neutron Scattering Research”, American Physical Society Global Physics Summit, Anaheim, CA (invited)

- "Neural Networks for Physicists", Data Science for Physicists Short Course, American Physical Society Global Physics Summit, Anaheim, CA (invited)
- "Applications of Artificial Intelligence to Neutron Scattering", Department of Physics, James Madison University (invited)
- "Applications of Artificial Intelligence to Neutron Scattering", Department of Physics, University of Buffalo (invited)
- "Applications of Artificial Intelligence to Neutron Scattering", Georgetown Data Science and Analytics Program (invited)

2024

- "Single Crystal Diffraction", Neutron X-Ray School (invited)
- "Applications of Artificial Intelligence to Neutron Scattering", Department of Physics, Johns Hopkins (invited)
- "Experimental Aspects of magnetic structure determination", Magnetic Structure School, Kennesaw State University (invited)
- "Incorporating Data Science in the Undergraduate Physics Curriculum", March Meeting of the American Physical Society (invited)
- "Unsupervised Learning for Physicists", Data Science for Physicists Tutorial, American Physical Society Global Physics Summit (invited)
- "Mapping the ICSD", March Meeting of the American Physical Society
- "Neutron Scattering and AI" (and panel), Neutron PI meeting, Rockville, MD (invited)
- "Semi and Self Supervised approaches to Space Group and Bravais Lattice Determination", Electronic Materials and Applications 2024, Denver, CO (invited)
- "Quantum Materials at NIST", Quantum Noir, Harvard University (invited)
- "Applications of Artificial Intelligence to Neutron Scattering", Department of Physics, Georgetown University (invited)
- "Semi and Self Supervised approaches to Space Group and Bravais Lattice Determination", Machine Learning Conference for X-Ray and Neutron-Based Experiments, FRM2, Garching, Germany
- "Applications of Artificial Intelligence to Neutron Scattering", Bakar Institute of Digital Materials for the Planet, Berkeley (invited)
- "Applications of Artificial Intelligence to Neutron Scattering", Department of Physics, University of Illinois (invited)
- "Adventures in Kagome Flat Band, $ErMn_6Sn_6$", Telluride Workshop, Telluride, CO (invited)
- "Applications of AI to crystal structure determination from powder diffraction data", European Crystallography Meeting/European Powder Diffraction meeting, Padova, Italy

2023

- "Integrating Data Science Into the Undergraduate Physics Curriculum", National Society of Black Physicists, Knoxville, TN

- “Semi and Self Supervised approaches to Space Group and Bravais Lattice Determination”, Pittsburgh Diffraction Society, Pittsburgh (invited)
- “Semi and Self Supervised approaches to Space Group and Bravais Lattice Determination”, March meeting of the American Physical Society, Las Vegas
- “Applications of Artificial Intelligence to Neutron Scattering”, Department of Physics, North Carolina A&T (invited)
- “Semi and Self Supervised approaches to Space Group and Bravais Lattice Determination”, International Union of Crystallography, Melbourne, Australia
- “Semi and Self Supervised approaches to Space Group and Bravais Lattice Determination”, American Chemical Society, San Francisco (invited)
- “Semi and Self Supervised approaches to Space Group and Bravais Lattice Determination”, Strongly Correlated Electronic Systems, Incheon, Korea (invited)
- “Semi and Self Supervised approaches to Space Group and Bravais Lattice Determination”, American Crystallography Association, Baltimore
- “Single Crystal Diffraction”, Neutron X-Ray School (invited)
- “DEI Panel: Fermilab Collaboration” Virtual
- “Panelist: The Evolution of Data Science Conference”, Georgetown

2022

- “Applications of Artificial Intelligence to Neutron Scattering”, Department of Physics, University of Connecticut (invited)
- “Neutron Investigations of Mn(Bi,Sb)2Te4”, MRS Meeting, Honolulu
- “Using AI to Determine Space Group from Neutron Powder Diffraction Data”, ACNS Meeting, Boulder
- “Reinforcement Learning and Neutron Scattering”, APS March Meeting, Chicago
- “APS Inclusion, Diversity, and Equity Alliance (IDEA), Transforming the Culture of Physics”, APS March Meeting, Chicago (invited)
- “Inelastic Neutron Scattering”, APS March Meeting, Chicago (invited DMP Short Course Lecture)

2021

- Opening Plenary speaker, APS-IDEA workshop
- “ORNL Postdoc Career Panel”, Oak Ridge National Laboratory (invited)
- “Investigations of the first intrinsic topological insulator: MnBi2Te4”, University of Maryland, College Park (virtual) (invited)
- “Investigations of the first intrinsic topological insulator: MnBi2Te4”, UC Santa Cruz (virtual) (invited)
- “Reinforcement Learning to optimize crystal structure determination”, March Meeting of the American Physical Society, Nashville, TN (virtual)

2020

- "Investigations of the first intrinsic topological insulator: MnBi2Te4", Fayetteville State University (virtual) (invited)
- "DPOLY Short course Machine Learning for Polymer Physicists: Neural Networks", March Meeting of the American Physical Society, Denver, CO
- "Red Cross Fire Risk", JSM 2020 Virtual Conference
- "Experimental Aspects of magnetic structure determination", Virtual Workshop on Magnetic Structures, Oak Ridge National Laboratory (virtual)

2019

- "Neutron Investigations of the Multiferroic Skyrmion GaV4S8", March Meeting of the American Physical Society, Boston, MA
- "Neutron Investigations of the Multiferroic Skyrmion GaV4S8", Strongly Correlated Electronic Systems Conference, Okayama, Japan

2018

- "Neutron Investigations of Thin Film Ho2Ti2O7", March Meeting of the American Physical Society, Los Angeles, CA
- "Mechanical control of the band structure of MoTe2", Telluride, Telluride, Colorado
- "Using neutron scattering to understand solid state materials", Corning Research, Sunnyvale, CA (invited)
- "Neutron Investigations of the Multiferroic Skyrmion GaV4S8", International Conference on Magnetism, San Francisco, CA
- "Multiferroic LuFeO3: Adventures in Neutron Scattering", Howard University, Department of Chemistry (invited)

2017

- "Effects of Electric Field on the magnetic structure of multiferroic (Sm,Bi)FeO3 films", March Meeting of the American Physical Society, New Orleans, LA
- "Magnetic structures and dynamics of multiferroic systems obtained with neutron scattering", Mid-Atlantic Sectional Meeting, Newark, NJ (invited)
- "Multiferroic h-LuFeO3–adventures in Neutron Scattering", Pittsburgh Diffraction Society, Indiana, PA (invited)
- "Neutron Investigations of thin film Ho2Ti2O7", Quantum Materials Symposium 2017, Berlin, Germany

2016

- "Bayesian Library for the Analysis of Neutron Diffraction Data", March Meeting of the American Physical Society, Baltimore, MD

- “Bayesian Library for the Analysis of Neutron Diffraction Data (BLAND)”, American Conference on Neutron Scattering
- “Bayesian Library for the Analysis of Neutron Diffraction Data”, American Crystallography Association, Denver
- “Atomically engineered ferroic layers yield a room temperature magnetoelectric multiferroic”, Gordon Conference on Multiferroics, Lewiston, Maine
- “Atomically engineered ferroic layers yield a room temperature magnetoelectric multiferroic”, Quantum Materials Synthesis Conference, New York Academies of Science, New York
- Invited Lecturer, Magnetic Structure Determination from Neutron Data, Florida State University, Tallahassee, FL

2015

- “Neutron scattering investigations of bulk and thin film multiferroic LuFeO3”, International Conference on Magnetism, Barcelona, Spain
- “Neutron investigations of multiferroic LuFeO3”, American Crystallography Association, Philadelphia
- “The intrinsic magnetic structure and ordering of multiferroic h-LuFeO3 Films”, March Meeting of the American Physical Society, San Antonio, TX

2014

- “Neutron Investigations of Multiferroic LuFeO3”, Gordon Research Conference, Biddeford, ME
- “Neutron diffraction investigations of BiFeO3”, Polarized Neutrons for Condensed Matter Investigations, U. Sydney, Sydney, Australia (invited)
- Invited Lecturer, Magnetic Structure Determination from Neutron Data, Oak Ridge National Lab, Knoxville, TN
- “Neutron Investigations of Multiferroic LuFeO3 Thin Films”, American Conference on Neutron Scattering, Knoxville, TN
- “Neutron Investigations of Multiferroic LuFe1-xMnxO3”, March Meeting of the American Physical Society, Denver
- “Neutron Investigations of Multiferroic LuFe1-xMnxO3”, Magnetic North IV, Victoria, British Columbia, Canada

2013

- “Evolution of the magnetic structure in (Sm,Bi)FeO3 Thin Films”, March Meeting of the American Physical Society, Baltimore
- “Neutron Investigations of Multiferroic LuFeO3”, Magnetism and Magnetic Materials, Denver, CO
- “Neutron Diffraction Investigations of Electric-Field Control of Antiferromagnetic Domains in Epitaxial BiFeO3 Thin Films”, Fundamental Physics of Ferroelectrics and Related Materials, Ames, IA

2012

- Invited Lecturer, Magnetic Structure Determination from Neutron Data, Oak Ridge National Lab, Oak Ridge, TN
- “Neutron Diffraction Investigations of Magnetism in BiFeO3 Epitaxial Films”, American Conference on Neutron Scattering (invited)
- “Investigation of Electric Field Control Antiferromagnetic Domains in Epitaxial BiFeO3 Thin Films Using Neutron Diffraction”, March Meeting of the American Physical Society
- “Neutron Diffraction Investigations of Magnetism in BiFeO3 Epitaxial Films”, DAE Solid State Physics Symposium, IIT Bombay (invited)

2011

- “Neutron Diffraction Investigations of Magnetism in BiFeO3 Epitaxial Films”, MMM 2011, Desert Ridge, Arizona (invited)
- “Neutron diffraction investigations of the BiFeO3 thin films”, European Spallation Source, Lund, Sweden (invited)
- “The Renaissance of Multiferroics”, Missouri University Science Technology (invited)
- “The Renaissance of Multiferroics”, Howard University (invited)
- “The Magnetic Form factor of SrFe2As2”, March Meeting of the American Physical Society, Dallas, Texas

2010

- “The Magnetic Form Factor in SrFe2As2”, MMM, Atlanta, Georgia
- “Neutron investigations of BiFeO3 Films”, March Meeting of the American Physical Society, Portland, Oregon
- “Introduction to SPINAL (Spinwave Analyzer), A program for Calculating and Analyzing Spinwaves”, American Conference on Neutron Scattering, Ottawa, Canada
- “BiFeO3”, Flipper, International Workshop on Single Crystal Diffraction with Polarized Neutrons, Grenoble, France
- “BiFeO3”, Aspen Winter Conference on Fundamental Physics of Ferroelectrics, Aspen, Colorado

2009

- “Neutron Scattering Studies of the Fe-Based Superconductors”, National Conference of Black Physicists, Nashville (invited)

2008

- “Dimensional Crossover in ZnMn2O4”, March Meeting of the American Physical Society, New Orleans, Louisiana
- “Dimensional Crossover in ZnMn2O4”, ACNS, Santa Fe, New Mexico

- “BiFeO3”, PNCMI, Mito, Japan (invited)

2007

- “Elucidation on the effects of hydrostatic pressure on multiferroic, HoMn2O5”, March Meeting of the American Physical Society, Denver
- “A New Magnet, ZnMn2O4”, National Conference of Black Physicists, Boston (invited)

2006

- “The low dimensional magnet, ZnMn2O4”, March Meeting of the American Physical Society, Baltimore, Maryland

2005

- “The Giant Magnetoelectric, DyMn2O5”, March Meeting of the American Physical Society, Los Angeles, California

2004

- “The Giant Magnetoelectric, DyMn2O5”, American Conference on Neutron Scattering, College Park, Maryland
- “The Magnetic Structure of ZnCr2O4”, March Meeting of the American Physical Society, Montreal, Canada

2003

- “Frustration in Flatland Revisited”, Poster, Boulder Summer School for Condensed Matter Physics, Boulder, Colorado

2002

- “Magnetism of Sr(Ti,Co)O3”, Poster, Gordon Research Conference on Strongly Correlated Electrons, Maine
- “Magnetism of Sr(Ti,Co)O3”, March Meeting of the American Physical Society, Indianapolis, Indiana

2001

- “Site-vs-Bond disorder in the geometrically frustrated magnet, ZnCr2O4”, Poster, SCES2001, University of Michigan, Ann Arbor
- “Site-vs-Bond disorder in the geometrically frustrated magnet, ZnCr2O4”, March Meeting of the APS, Seattle, Washington

2000

- "The geometrically frustrated magnet, $ZnCr_2O_4$", Poster, Gordon Research Conference Strongly Correlated Electrons, New Hampshire

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Instruction Experience

- Conducted numerous summer schools at the NCNR
- Summer High School Intern Program (SHIP) mentor for high school students
- Summer Undergraduate Research Fellowship (SURF) mentor for undergraduate students
- Outreach through Adventures in Science
- Part time lecturer Physics 205, Rutgers (Summer 2003)
- Curriculum development and teaching of 6th and 9th graders through the NSF GK-12 fellowship (1999-2001)
- Part time lecturer Physics 203, Rutgers (Summer 1999)
- Part time lecturer Physics of Sound, Rutgers (Spring 1999)
- Part time lecturer Physics 204, Rutgers (Spring 1999)
- Part time lecturer Honors Modern Physics, Rutgers (Fall 1998)
- Part time lecturer Physics 204, Rutgers (Summer 1998)
- Part time lecturer Physics 203, Rutgers (Summer 1998)

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